

BST

Class Lecture-6

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Binary tree

Binary tree is

- a root
- left subtree (*maybe empty*)
- right subtree (*maybe empty*)

Binary Search Trees

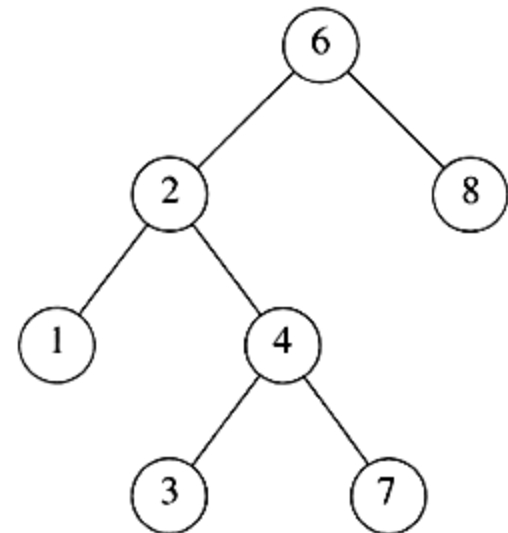
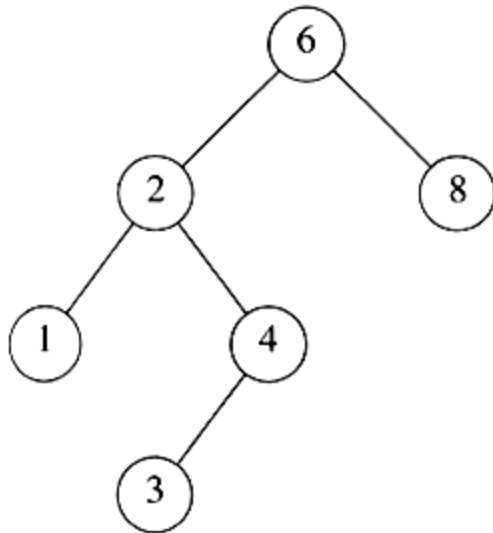
Binary **search** tree property

For every node X ,

- 1) All the keys in its left sub-tree are smaller than the key value in X ,
- 2) All the keys in its right sub-tree are larger than the key value in X

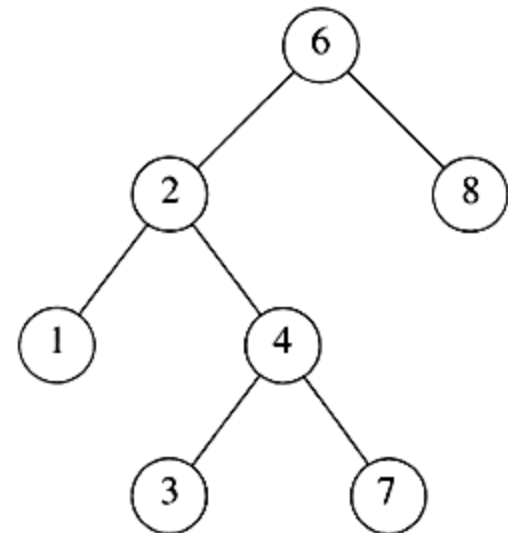
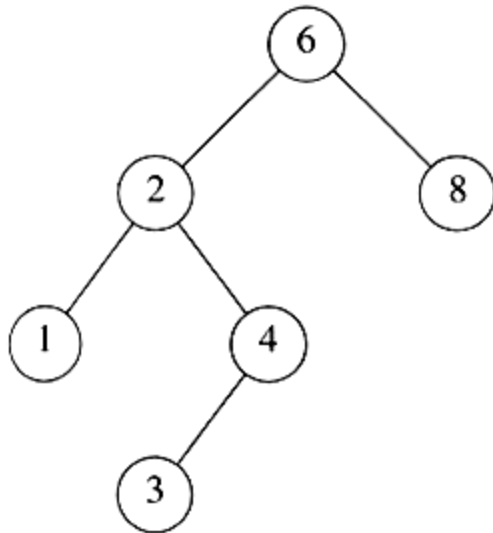
Binary Search Trees

Determine Whether the given trees are Binary Search Trees or not!



Binary Search Trees

Determine Whether the given trees are Binary Search Trees or not!



BST Construction Cont.

30 15 22 60 45 17 75 7 27

Construct BST with the above elements.

BST Construction Cont.

30 15 22 60 45 17 75 7 27

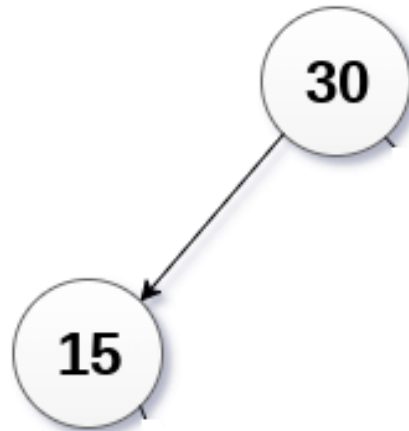
30



BST Construction Cont.

30 15 22 60 45 17 75 7 27

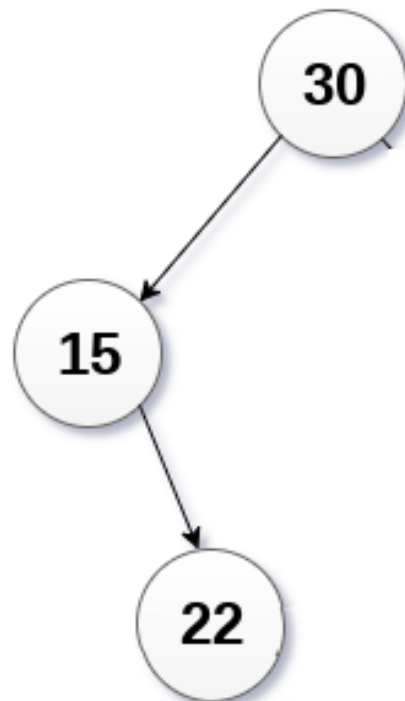
30 15



BST Construction Cont.

30 15 22 60 45 17 75 7 27

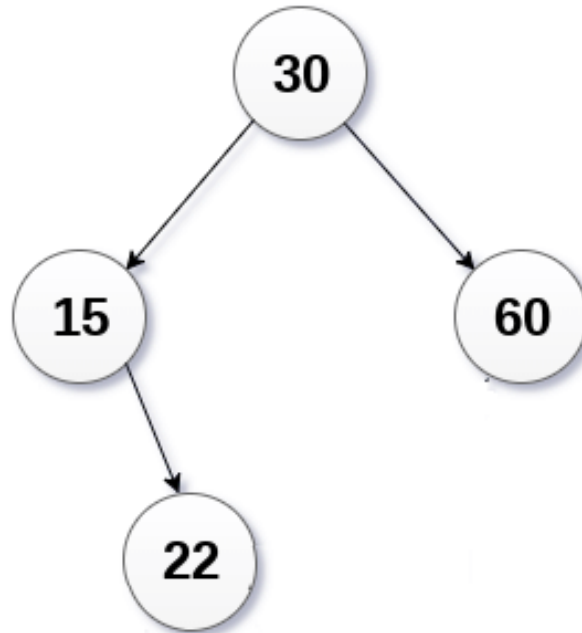
30 15 **22**



BST Construction Cont.

30 15 22 60 45 17 75 7 27

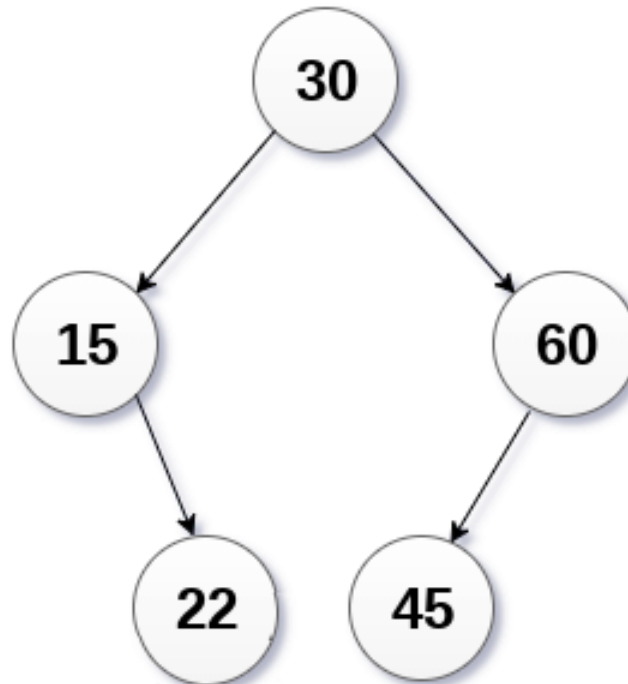
30 15 22 **60**



BST Construction Cont.

30 15 22 60 45 17 75 7 27

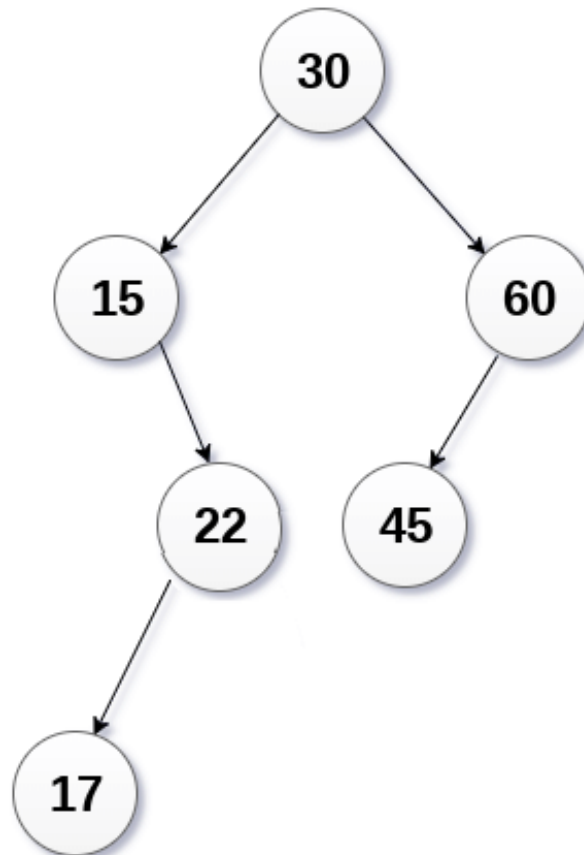
30 15 22 60 45



BST Construction Cont.

30 15 22 60 45 17 75 7 27

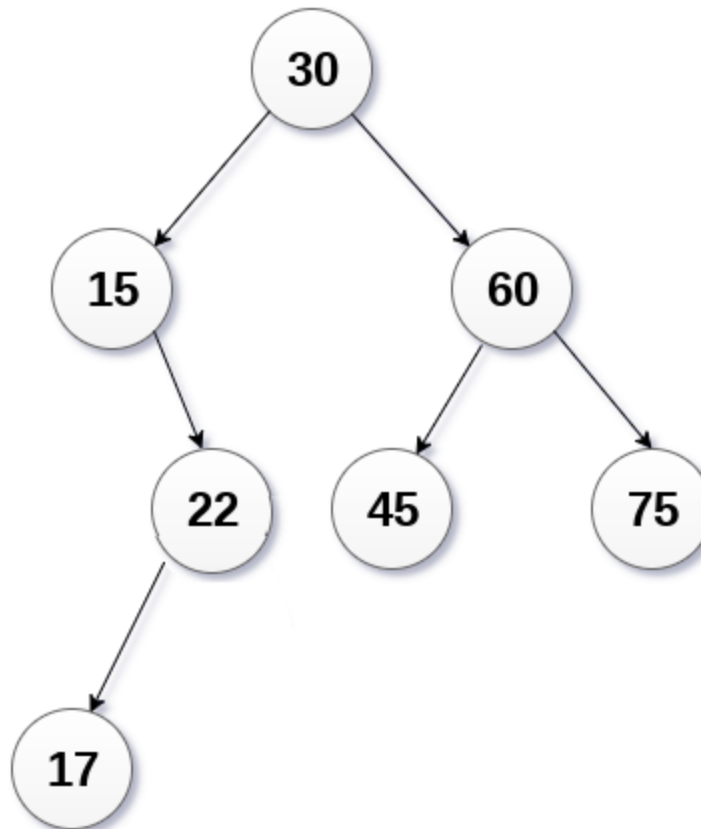
30 15 22 60 45 17



BST Construction Cont.

30 15 22 60 45 17 75 7 27

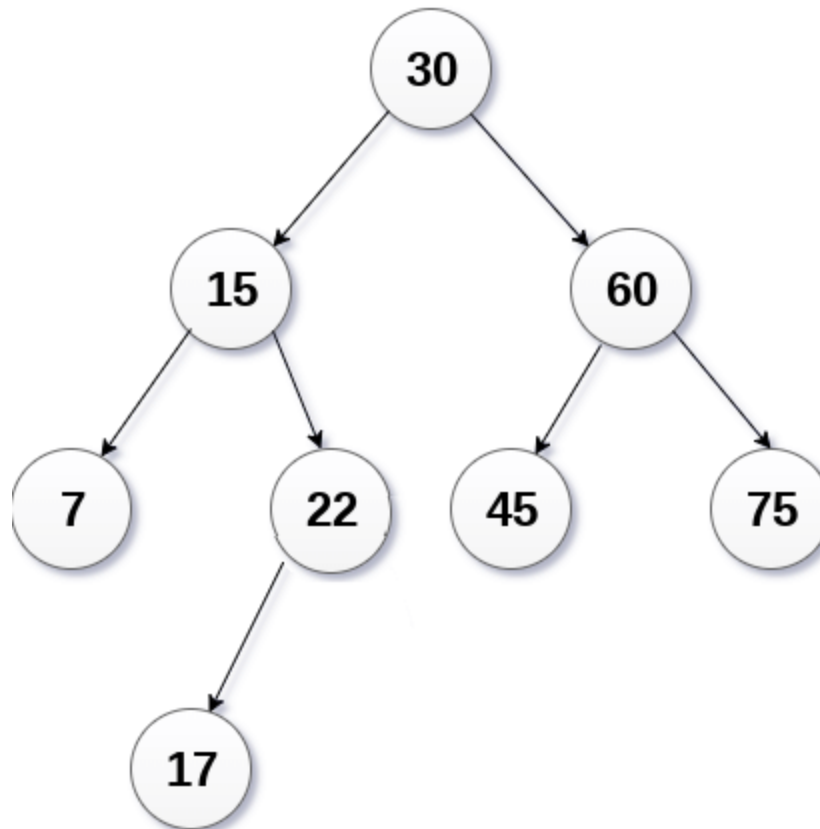
30 15 22 60 45 17 **75**



BST Construction Cont.

30 15 22 60 45 17 75 7 27

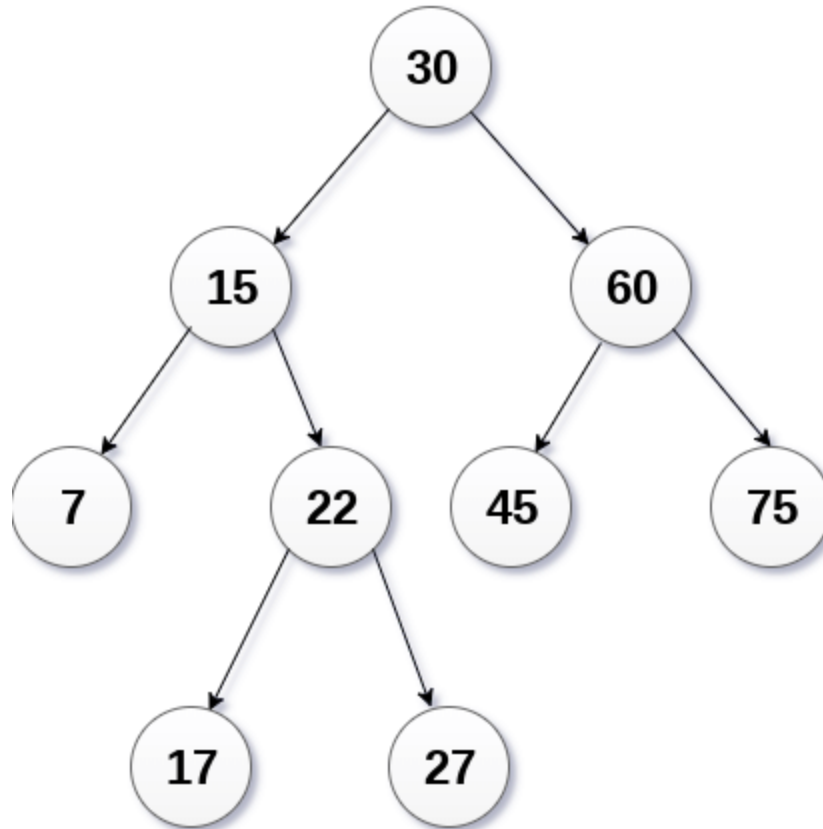
30 15 22 60 45 17 75 **7**



BST Construction Cont.

30 15 22 60 45 17 75 7 27

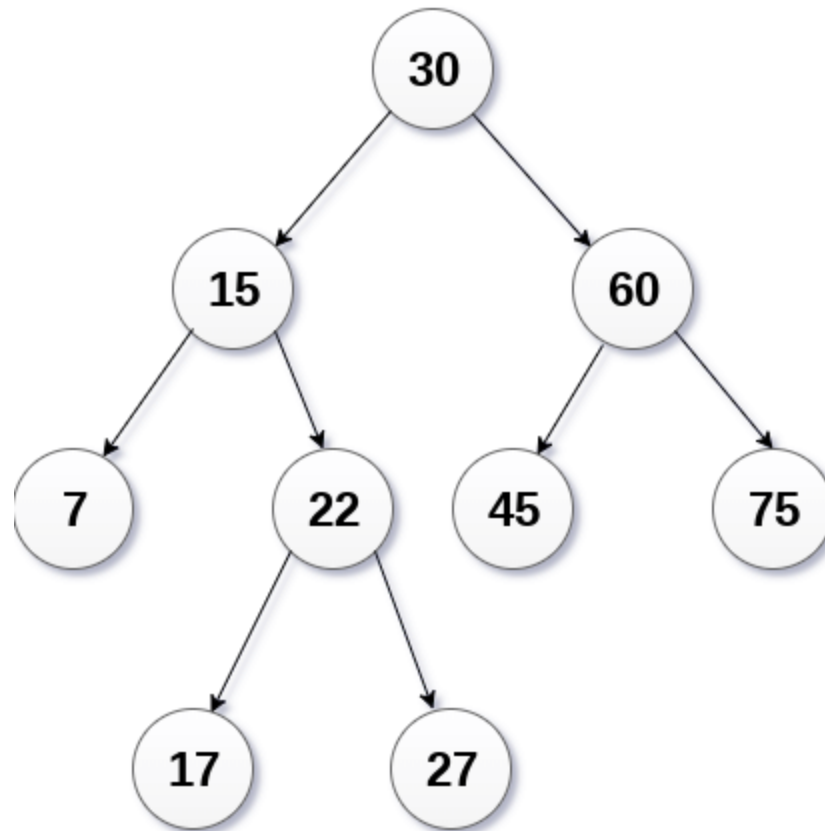
Find 27 in the array



BST Construction Cont.

30 15 22 60 45 17 75 7 27

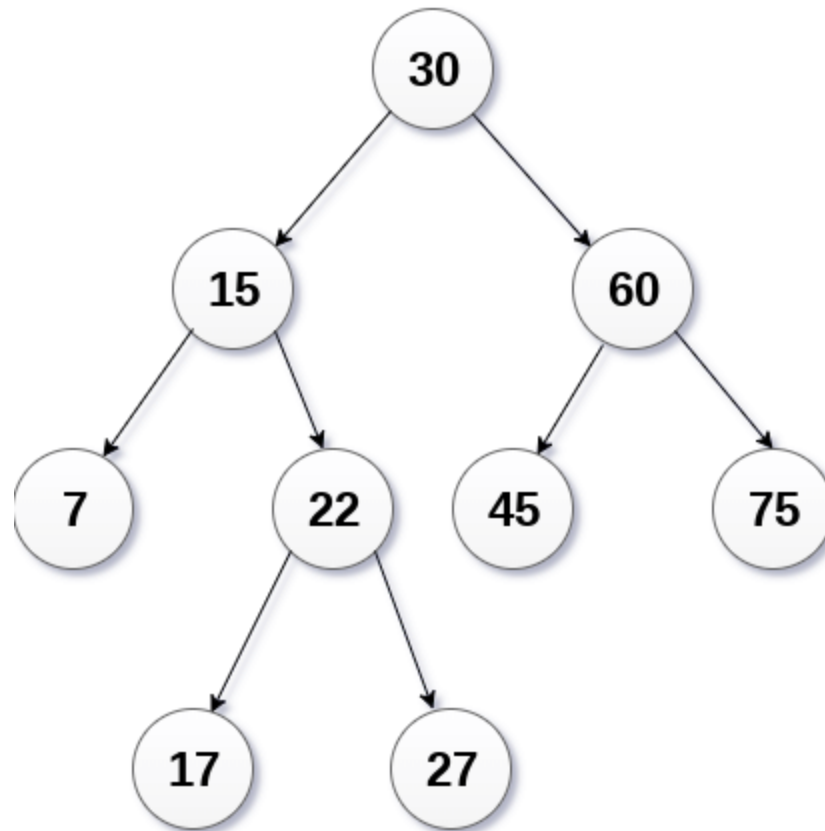
Find 25 in the array



BST Construction Cont.

30 15 22 60 45 17 75 7 27

30 15 22 60 45 17 75 7 **27**



Thank You